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Exploring the Use of Landmarks to Aid Pedestrian Wayfinding

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Featured Application: Using photographs of conspicuous landmarks in mobile maps can enhance the spatial information presented to pedestrians when wayfinding.

Abstract: When wayfinding with mobile maps, the acquisition of spatial knowledge can be limited by relying on automated route instructions and the small map on the screen. Repeating a route without support from navigation aids may be challenging as the user has been focused on their mobile phone and not the surroundings. According to theories of spatial knowledge acquisition, landmarks are environmental elements important in the development of such knowledge. This study examines how different navigation aids impact spatial knowledge, with a particular focus on landmarks. Participants navigated a route using one of three aids: a booklet of sequential landmarks, a booklet of Google Maps screenshots, or a paper map. The landmarks were selected based on their conspicuity and strategic placement along the route. Thirty participants completed tasks assessing their spatial knowledge of the route and surrounding area after walking it. This study, divided into three phases, compared the effectiveness of each navigation aid, highlighting the effectiveness of landmarks in wayfinding. Results indicate that landmarks significantly enhance wayfinding, especially for pedestrians navigating short, pre-set routes without a map. This suggests that incorporating landmarks into mobile maps could improve on-screen spatial information.

Keywords: landmarks; spatial knowledge; mobile maps; wayfinding; navigation

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1. Introduction

Landmarks have distinctive properties that can assist people in finding their way [1–3]. This study's major goal was to determine if photographs of landmarks can be used by pedestrians to follow a route to aid the development of spatial knowledge of surroundings. If this is the case, subsequent research could then explore their use in mobile maps as problems exist when wayfinding with such maps—users use a very small, constrained map of a region and the route is generated by an automated system, which entails little thinking and planning on behalf of the user [4–6].

Little research has been conducted where landmarks have been the principal navigational aid used to follow a route, exceptions being [7,8]. We investigated the extent to which a static landmark-based wayfinding system may facilitate pedestrian wayfinding and the acquisition of landmark, route, and survey knowledge compared with a static map-based system based on Google Maps and a conventional static paper map. The length of the chosen route that participants were asked to take was similar to that reported in another relevant research [2,9,10]. The route was located in a landmark-rich region that included several turns comparable to those of those other studies. Different navigational aids were provided to three participant groups to accomplish the route-following tasks. Sequential images of the landmarks along the route were shown to the first group. The second group used a small book of screenshots of the route as depicted in Google Maps, while the third group relied on a paper map to determine their own paths. A printed booklet version of the route in Google Maps was used by the second group, as it was

thought that real mobile maps would offer an unfair advantage to participants using this aid. Therefore, all three wayfinding aids were paper-based. Typically, GPS-based mobile maps can provide precise location information, step-by-step directions, and augmented reality overlays aligned with the user's perspective. By leveraging these sensors, users can effortlessly navigate unfamiliar environments, find the nearest points of interest, and seamlessly explore their surroundings with confidence.

2. Literature

With the arrival of the smartphone variant of the mobile phone in 2007, pedestrian wayfinding was significantly transformed. Smartphones provide accurate detection and interpretation of users' movements and orientation in real-time by using their inbuilt sensor-based systems, such as the Global Positioning System (GPS), an accelerometer, a gyroscope, and a magnetometer. Nevertheless, GPS is not a perfect system—mobile maps can be drawn or labeled erroneously, so even if the above sensors perform correctly, an incorrect map can still be shown [4]. Aside from occasional map-making errors encountered in mobile maps, two key issues emerge when wayfinding with these systems, namely, (i) the keyhole problem [5,6] and (ii) problems with automated systems.

The keyhole problem refers to when elements of interface design restrict the viewing of information on-screen—hindering interface usability. Web browser windows that show cropped portions of images that require a user to resize or scroll the page to display the entire image are an example of this problem. Another example is that some text fields on a website can only accommodate a small section of long words that exceed the field size. In the context of mobile maps, when viewing a map of a large geographic area on a small mobile device, only a small section of the larger map is readily viewable. That is, the user can only perceive a 'keyhole' view of the map. This is shown in Figure 1 below.

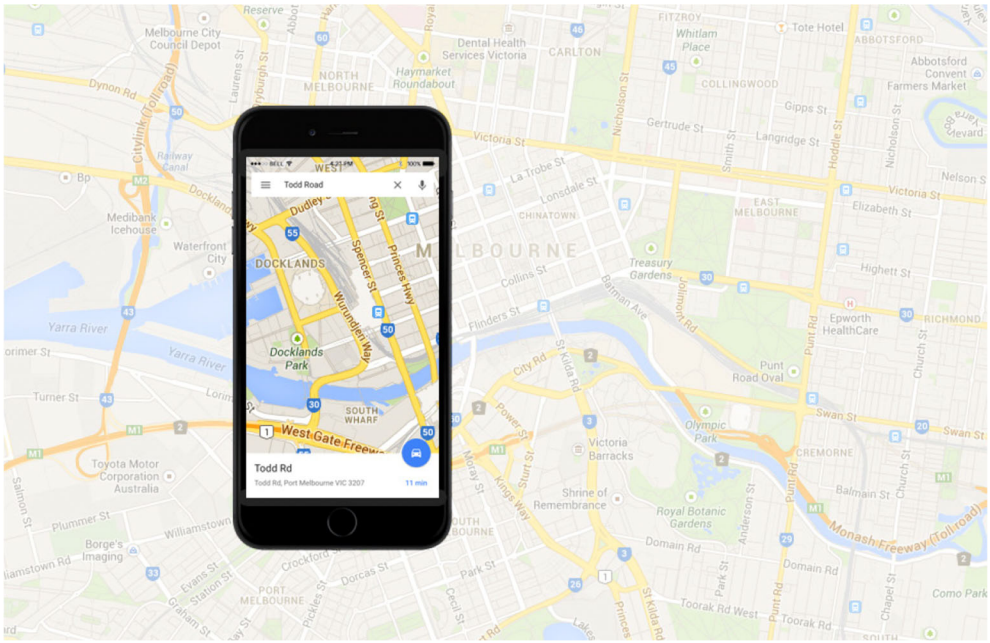


Figure 1. The keyhole problem demonstrated. Map data © 2015 Google.

There are three interrelated aspects to this problem with mobile maps: (i) the small size and the unstable nature of the map presented on-screen [11]; (ii) the high interaction cost of manipulating the interface to view the map proper; and the related (iii) view-change cost, where shifts between global and local views need to be understood [12].

Map instability occurs due to the GPS-based mobile maps moving to correspond with the user's position or viewing direction and the zoom-in, zoom-out aspect of mobile maps. Interaction cost refers to the number of screen-based interactions a person has to complete

in order to complete a task within an interface. A high interaction cost complicates the task [13]. It is possible that contemporary mobile maps may have a lower interaction cost than earlier versions, such as what was used in earlier studies (e.g., [9,11]). Before route directions zoomed-in to upcoming turns in turn-by-turn mode, users needed to zoom in and out of the map much more frequently to increase the details of the maps.

The second key problem when using mobile maps for wayfinding is relying on the automated route directions. Automation is often described as a labor-saving process that can extend or aid human capability. However, automation can also be a "thinking–saving" process where users implicitly trust a system that may fail—with users lacking the ability to deal with the situation. In the case of automated mobile maps, should they malfunction, the wayfinder may not be aware of the current location and might become disoriented or lost. However, when using a paper map to plan a route, the user engages in several cognitive processes to make sense of the map in relation to the surroundings. Lobben [14] outlines several theories underlying those processes. Three of those are of interest here, namely map rotation, self-location, and visualization, all of which highlight the differences between using paper maps and mobile maps. With map rotation, cartographic tradition prescribes that a "north-up" orientation be used when creating paper maps. This means that in order to relate the map to the environment, the map is rotated, either physically or mentally, to align with the surroundings [15]. This transforms the map from being allocentric, where the spatial relations between objects in a region are described, to egocentric, where the spatial relations between the map user and objects are described [16]. Self-location occurs when the map user is able to locate their current location on the map; this process happens at the commencement of a wayfinding task. The physical surroundings, landmarks in particular, are matched to their location on the map, enabling orientation to occur. This is a problem-solving activity [14]. Visualization, an ongoing cognitive process when wayfinding, refers to another form of map and environment interaction whereby the map's two-dimensional representation of the surroundings is matched to the three-dimensional real-world environment. For Lobben [14], unlike matching the environment to the map, as with self-location, visualization involves matching the map to the environment while moving through it. The "mental representations [thus] formed and the patterns people see are closely linked to expectations they bring to a given situation" [17] (p. 101). Matching the map to the surroundings gives the map-user the ability to predict upcoming spatial configurations in the real world, which is continually repeated as wayfinding continues.

Automation is defined as "the execution by a machine agent (usually a computer) of a function previously carried out by a human" [18] (p. 231). When using mobile maps, two theoretical spatial procedures are irrelevant: map rotation and self-location are handled by the automated system [14]. This means that the cognitive effort of paper map rotation is not required as the mobile map system provides a heads-up map, also known as a track-up map. At the simple touch of a button, the map region located directly in front of the user is displayed, which may or may not be north [19]. The GPS automatically locates the user, which removes the need for self-location. The third of Lobben's [14] spatial processes, namely visualization, still occurs with mobile maps. Visualization matches the map to the surroundings as the user moves through the region. However, the amount of visualization provided via a mobile map is probably less than what occurs with a paper map. The small size of the map on mobile phones reduces one's ability effectively to match the map to the surroundings [9,11], which can result in the so-called keyhole problem.

Landmarks present an opportunity for use in the constrained interface of mobile maps as they have been proven to aid our understanding of routes and the surrounding environment [20–23]. Research indicates that landmarks are vital environmental elements used when wayfinding. According to some researchers, we use landmarks when we follow a route, in particular at intersections [24], as landmarks can usefully provide orientation in unfamiliar environments [20]. We also tend to use them when giving verbal route instructions [21,22] and when drawing sketch maps of a route [23].

Currently, landmarks do not feature in digital mapping systems. With contemporary mobile maps, it appears that technology may exist for depicting landmarks on screen. Here WeGo Map Version 2017), in particular, appear to be the most advanced in this regard. As Figure 2 shows, the turn-by-turn instructions in Here WeGo Maps use 3D renderings of buildings, and while these buildings may be landmarks in a general sense (i.e., they are prominent buildings/landmarks within cities), they are not used as decision points or confirmation points in route instructions. At present, no technology exists that successfully identifies landmarks based on salience or appropriateness for route instructions and integrates them into a commercially available mapping system [25,26].

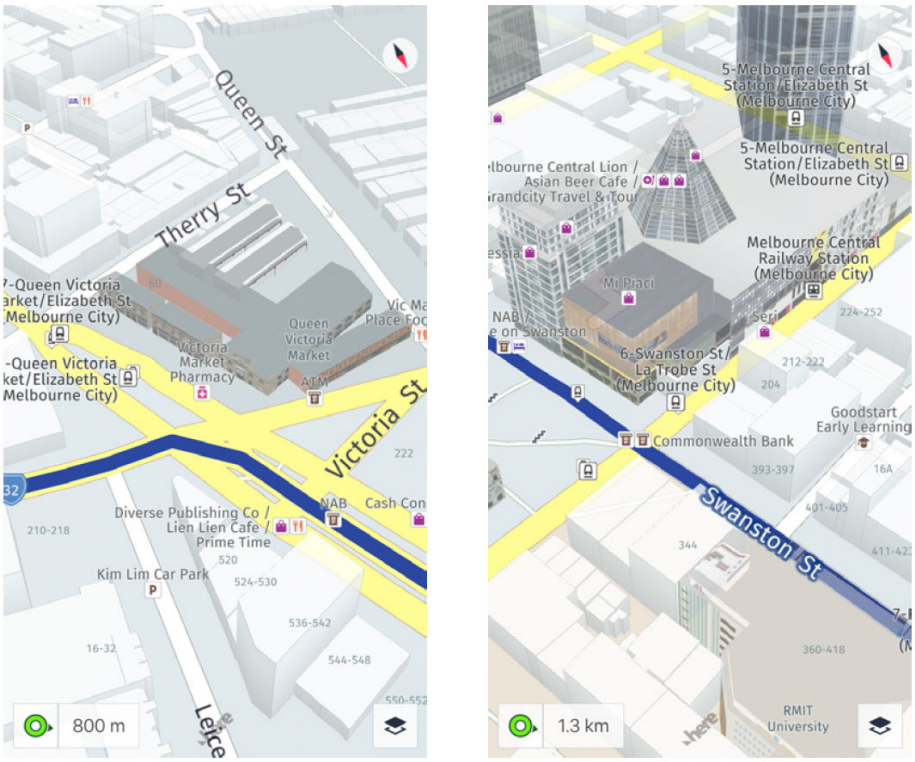


Figure 2. Examples of 3D buildings used in Here WeGo Maps. There is potential for these to indicate actual landmarks. © 1987–2017 HERE.

Deakin [1] investigated if landmarks presented in a street map would enhance the available spatial information and aid route following. To address this, participants were assigned to one of three participant groups, all of which used a conventional street map. Twenty salient landmarks, all located at intersections, were used. Various geometric symbols depicting the fifteen landmark types (i.e., post office, library, and park) were added to the map given to one participant group; sketches depicting the landmarks were added to the second group's map, and no supplemental landmarks were added to the map given to the third group. Deakin's participants interacted with a slide show of the route where they were to choose the correct direction at each intersection to reach the same destination across town—differing from the real-world wayfinding scenario of this research. Participants could refer to their supplied map. The number of errors made and the speed of the participant in making decisions at each decision point were noted.

The two participant groups whose maps included landmarks made significantly fewer turn errors than the no-landmark group, but their decision-making was significantly slower. Deakin [1] interpreted this as suggesting that the landmarks aided the decision-making process, in particular at complex intersections, as participants claimed that the more realistic sketches helped most at these points. However, it was observed that the users of the symbol landmarks had to refer to a key to understand what each symbol meant, slowing their response, and interpreting some of the sketches may also have slowed progress. In this

study, color photographs of landmarks were used as they clearly depict their subject matter, needing no interpretation of the images.

The landmarks selected in this research adhered to a model of landmark salience where conspicuity and prominent location were key selection criteria [27]—similar to the approach used by Deakin [1]. Research that investigates wayfinding with landmarks without linking them to concepts of landmark salience (e.g., [28]) creates uncertainty over the 'landmarks' term, leaving it open to subjective interpretation.

3. Method

3.1. Participants

Thirty participants of non-visual communication professions/tertiary courses (15 females and 15 males) were recruited via an online classified advertising website and Facebook. Participants were aged 18–50 years (M = 28.3 years, SD = 8.99); all were proficient in English, and all were unfamiliar with the geographical test region. Participants were tested in individual sessions taking approximately one hour, not including travel time to the study's starting point. At the end, they were paid AUD 40 each in department store gift cards and excused. This study included three phases with three groups. In Phase 1, participants in two of the three groups were asked to walk the same pre-set route—which was the fastest route through the region as determined by Google Maps. Due to the fact that one group had been shown photographs of the conspicuous landmarks along the route (the Landmark [LM] Group) as shown in Figure 3, the route-following task was deemed easier for that group than for the group who relied on Google Maps (the GM Group), whose task was also to follow the pre-set route. Hypothesis 1 therefore predicted that the LM Group would outperform the GM Group by making fewer wayfinding errors and fewer hesitations. As participants in the paper map group (the PM Group) were to choose their own route, their task differed from that of the two other groups. Therefore, their data were excluded from the analysis of this task. This latter group was included to see what route they would take to the destination and if this route would be similar to the (pre-set) fastest route as computed by Google Maps—as used by the LM and GM Groups. The time to travel the route was not considered in this study as the researchers thought that accuracy in navigation—not taking wrong turns and being able to find your way if a wrong turn was taken—was more important than travel speed.



Figure 3. Two distinctively designed buildings used on the route as landmarks by the LM Group.

In Phase 2, participants in the LM and GM groups were asked to draw the route on an unlabeled paper map. Because it was reasonable to expect participants in the LM Group to primarily focus on the landmarks, this should negatively affect their ability to draw the route on an unlabeled paper map compared with participants in the GM Group. Accordingly, hypothesis 2 predicted that the route drawn by the GM Group would be more

accurate than the route drawn by participants in the LM Group. As the PM Group drew their own routes, their data were again excluded from the analysis in this phase.

In Phase 3, all participants were asked to place all the landmarks encountered on an unlabeled paper map of the region. Assuming that participants in the LM Group would attend more landmarks than participants in the GM and PM Groups, hypothesis 3 predicted that the LM Group participants would locate more landmarks correctly than the GM Group and the PM Group, who walked their own routes. The likelihood of noticing the on-route landmarks was clearly higher than that of noticing distant off-route landmarks (often used for orientation and creating a cognitive map of the region). Accordingly, hypothesis 4 predicted that all three participant groups would locate more on-route landmarks than off-route landmarks. Finally, as the LM Group's navigation aid had implicitly encouraged them to focus on landmarks in Phase 1, hypothesis 5 predicted that the LM Group would locate more off-route landmarks correctly than the participants in both the GM and the PM Groups. Table 1 below summarizes the hypotheses of this study.

Table 1. The five hypotheses for this study.

Hypothesis	Prediction	Expected Outcome
1	The LM Group will outperform the GM Group in terms of fewer errors and hesitations	Main effect for groups
2	The GM Group's route drawings will be more accurate than those of the LM Group	Main effect for groups
3	LM Group participants will locate more landmarks correctly than either the GM and the PM Groups	Main effect for groups
4	All participants will locate more on-route than off-route landmarks correctly	Main effect for landmarks (on/off route)
5	LM Group participants will locate more off-route landmarks correctly than the GM and PM Groups	Main effect for groups on off-route landmarks

3.2. Materials

An information statement and an informed consent form were prepared along with a group-specific booklet of 12 pages for the LM and GM participant groups, respectively, and the PM Group was given an A4 printed Google Map (desktop version—Google, Mountain View, CA, USA) of the region showing street and place names. Also, prepared was the Santa Barbara Sense of Direction (SBSOD) scale [29], which contains 15 statements, eight of which are negatively phrased. These were converted to positive statements, such that higher scores indicated better self-rated spatial skills. Each statement was scored on a 7-point Likert scale, in which a score of 1 represented maximum disagreement and a score of 7 indicated maximum agreement with the statement. The LM Group's booklet contained task instructions, a visual explanation of how the landmark system worked, and photographs of 16 sequential on-route landmarks. The landmarks were classified as decision points, found at intersections, and confirmation points confirming the route was being correctly followed. Accompanying the decision points were brief written route instructions with directional arrows, as shown in Figure 4. Decision points were presented in a larger size than confirmation points. Aside from the two introductory pages and the visual explanation page, the remaining nine pages each contained between zero and two confirmation points and a decision point. In total, there were seven confirmation points and nine decision points. The word landmark was not used in the booklet; the term "photo sequence" was used instead.

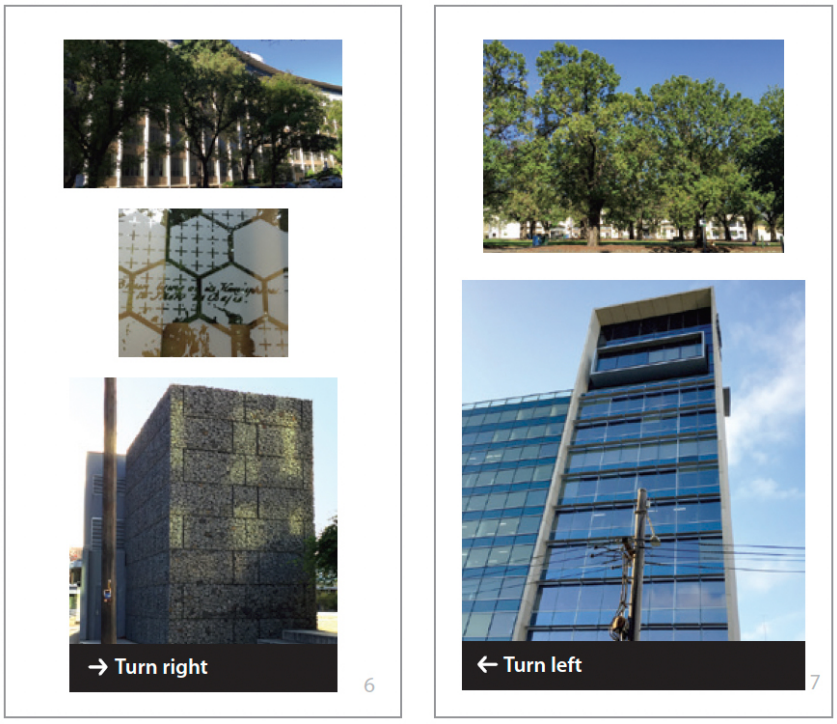


Figure 4. Pages from the LM Group's booklet.

The GM booklet contained a page of task instructions, a page depicting a screenshot of the overview map of the entire route shown in context, and a further ten pages of individual screenshots depicting one of the eight route segments as shown in Figure 5. Some segments were shown from two viewpoints—as in the Google Maps (v.2016) mobile application. As some of the street names were unreadable when presented on paper, these were retouched to enhance readability. Pages in the printed booklets matched the screen size of an Apple iPhone (Apple Inc., Melbourne, Australia) (104 mm × 158 mm). The PM Group used a paper map of the region containing street and place names with the start point and destination marked.

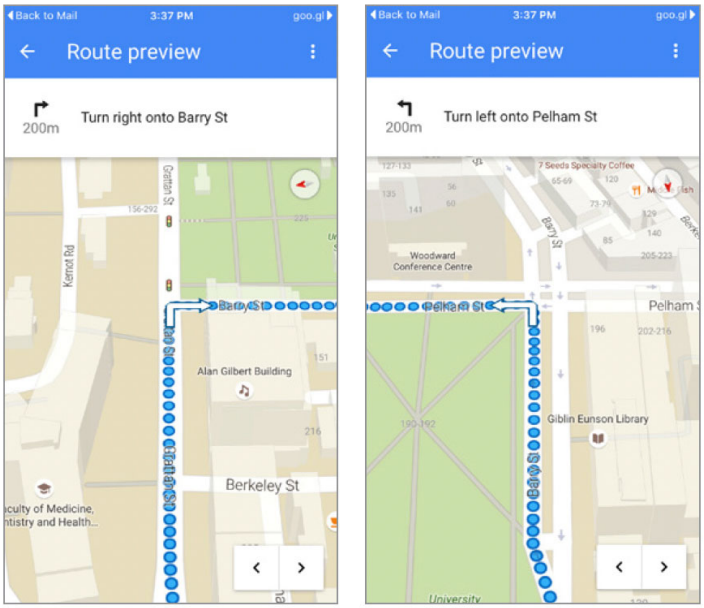


Figure 5. The corresponding pages from the GM Group's booklet. Map data © 2016 Google.

An unlabeled A3 printed map of the entire region was used in Phases 2 and 3. Also used in Phase 3 were three pages containing numbered photographs of eight 'on' and eight 'off' route landmarks. The eight on-route landmarks were selected from the 16 presented in the LM Group's booklet. Figure 6 depicts some of these landmarks, and Figure 7 outlines the connection between the LM Group's landmarks used in Phase 1 and those used in Phase 3. Figure 8 summarizes the materials used in each of the three study phases.

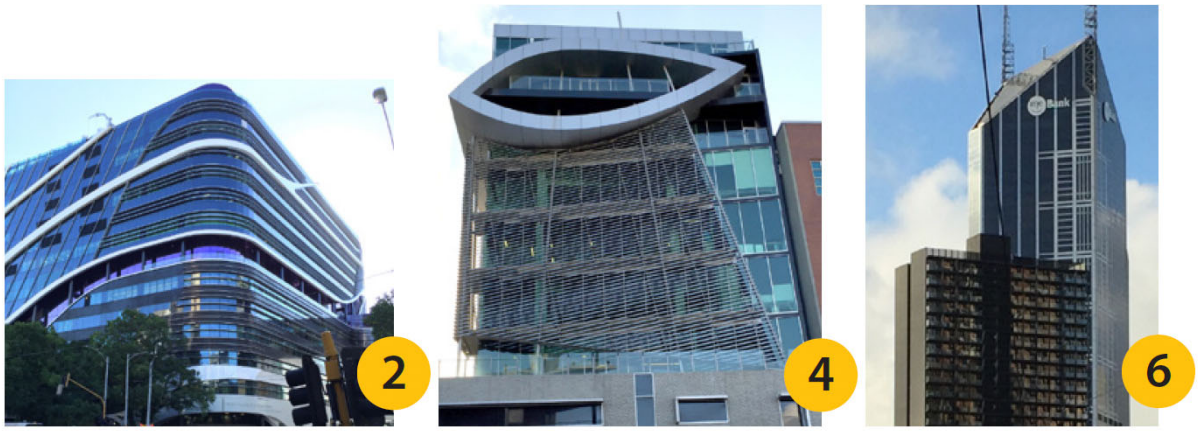


Figure 6. A selection of the on- and off-route landmarks used in Phase 3. Landmarks 2 and 4 were on-route, while landmark 6 was off-route.

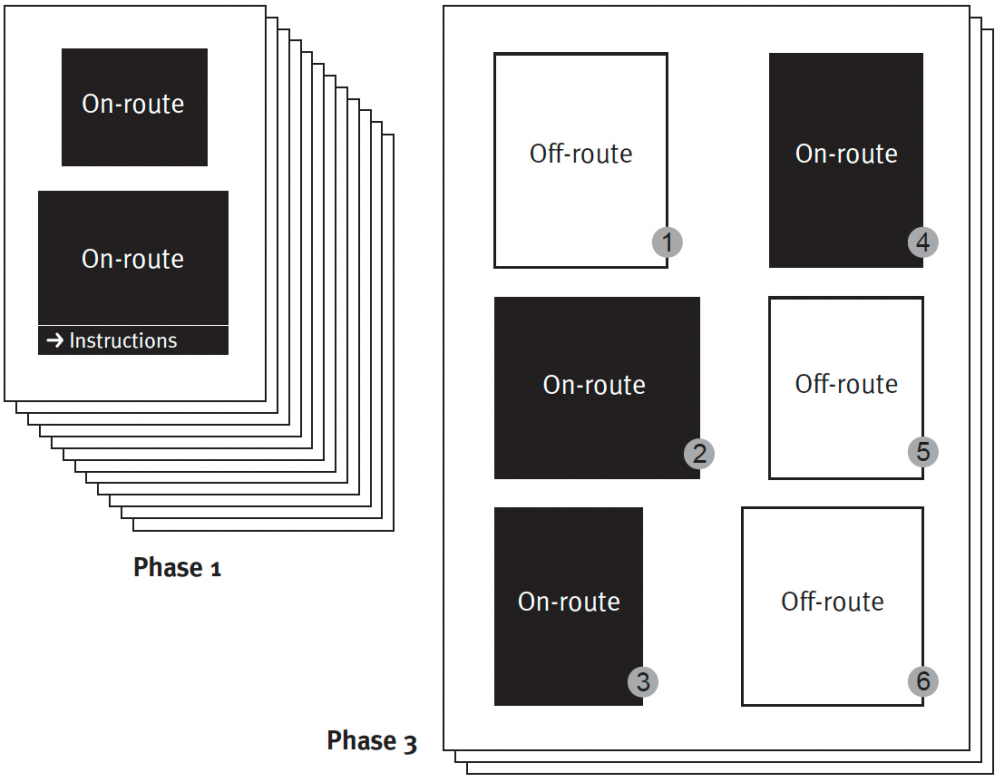


Figure 7. The materials that used landmarks. Phase 1: The LM Group's booklet contained 16 on-route landmarks shown in sequential order. Phase 3: 16 photographs of landmarks (eight on-route, eight off-route) displayed in random sequence—these were matched to their location on the unlabeled paper map. The random numbers 1–6 identify the photographs of landmarks that were then written on the unlabeled paper map.

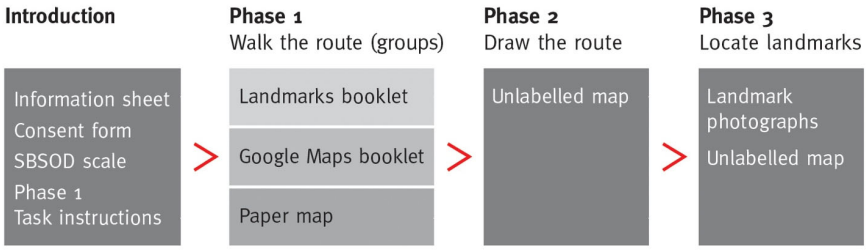


Figure 8. Overview of the materials used.

3.3. Design

This study used a 'between-subjects' design in which participants were randomly assigned to one of the three groups: the LM, GM, or PM Group. In Phase 1, participants in the LM and GM Groups followed the pre-set route guided by their respective booklets. The LM Group's route consisted of eight segments with an on-route landmark assigned to each turn (a decision point) and almost every segment (a confirmation point). Sixteen on-route landmarks were shown in the LM booklet. The GM Group's booklet provided Google Maps screenshots of the route, showing each of the eight route segments. Figure 9 depicts the locations of the 16 on-route landmarks shown to the LM Group in their booklet and the route used by the LM and GM Groups. Participants in the PM Group were given the A4 map and instructed to take the quickest route to the destination. All participants were timed and instructed to walk at a comfortable pace. Route-following accuracy was to be prioritized over travel speed. Timing was paused when stopped at traffic lights. The researcher provided instructions directing LM and GM Group participants back to the last decision point/intersection to help them re-orientate and attempt the route again if they made an error that took them off the route for 20 s. Hesitations of 20 s or more by the LM or GM Group participants were noted. Participants in the PM Group were asked to give a quick overview of their planned route before starting their walk, enabling the researcher to know if/when a participant had deviated from their planned route. All walks were completed between 10:30 a.m. and 4:00 p.m. during winter. Timing was captured on an Apple iPhone stopwatch, and the researcher relied on his substantial local knowledge of the test area to guide participants back to their route if they strayed from it.

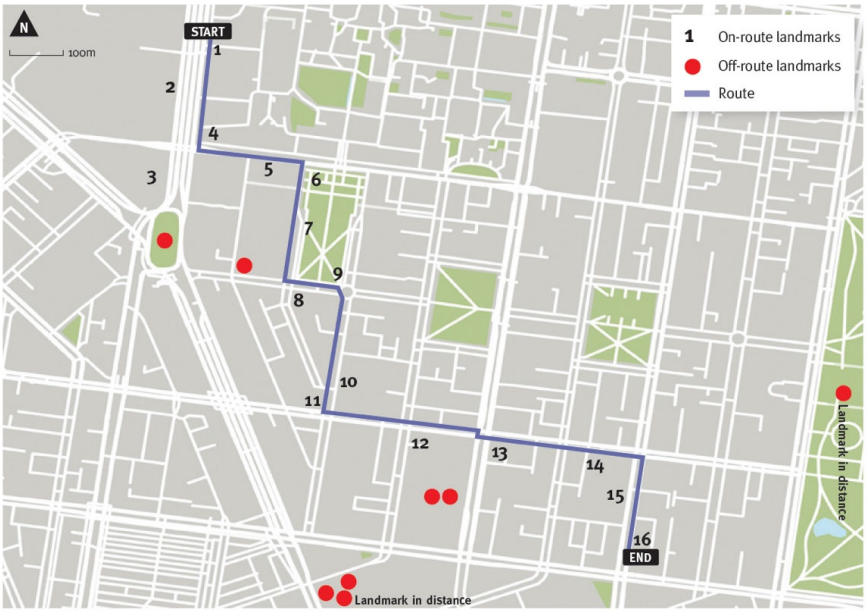


Figure 9. The route followed by the LM and the GM Groups in Phase 1, with the 16 on-route landmarks used in the LM Group's booklet indicated. Also shown are the off-route landmarks used in Phase 3. Map data © Mapbox, © OpenStreetMap.

In Phase 2, all participants drew the route they had traveled as accurately as possible on the unlabeled A3 paper map that depicted the start point and the destination. In Phase 3, participants attempted to locate 16 numbered landmarks (eight on-route; eight off-route—on another identical unlabeled A3 paper map). The photographs of landmarks were arranged in random order with the numbers applied in sequential order. Eight of the on-route landmarks were used in the LM Group's booklet in Phase 1, and the eight off-route landmarks were not. The same order was used for all participants. The off-route landmarks were located between 100 and 500 m from the route. Participants located the landmarks by writing the number of each photograph showing the relevant landmark on the map. A landmark located within 100 m of its actual position was regarded as correct. Figure 10 summarizes the study design.

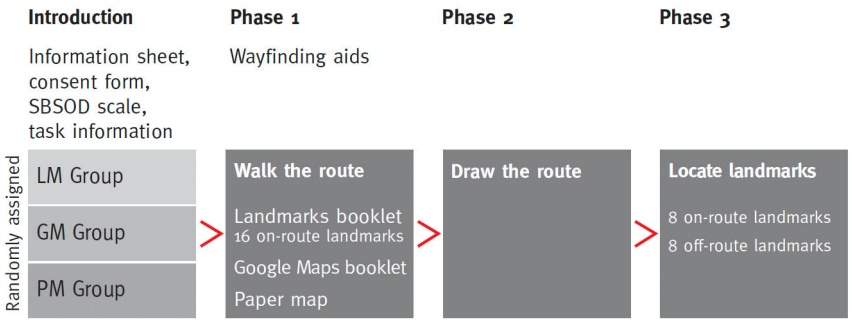


Figure 10. Summary of the study design.

3.4. Procedure

Each participant was met at the start point. Upon reading and signing the informed consent form and completing the SBSOD scale, each participant was given the relevant booklet (LM, GM Groups) or map (PM Group) and asked to read the task instructions on the navigation aids. The LM Group's task instructions outlined the function of the two landmark photographs on the pages of their booklet (one photograph being a decision point, or turn point, the other a confirmation point) and that the accompanying route instructions were to be followed. The GM Group's task instructions were to follow the printed screenshots of the route instructions from Google Maps, and the PM Group was asked to walk the quickest route between the start-point and destination indicated on a paper A4 map of the region. Each participant was given an unlabeled A3 map on which to draw the route (or as much of it as they could). The route could be drawn from start to end, in reverse, or in piecemeal fashion. For Phase 3, participants were given a clean paper map and the set of three A4 sheets containing photographs of landmarks, and participants were asked to write the number of each landmark on its location on the map with start and end points indicated. Upon completion of this task, participants were invited to ask questions or make any comments on the study.

4. Results

In this study, all analyses of results were tested for violation of assumptions underlying the Analysis of Variance (ANOVA), the formal analysis techniques applied. These assumptions are as follows: (i) the data do not contain significant outliers, as determined by assessment of a boxplot; (ii) the data shall be normally distributed, as tested by the Shapiro–Wilk test; and (iii) homogeneity of variances exists in the data, as determined by Levene's test of equality of variances. Efforts were made, when required, to tweak the data to match these assumptions via data transformations. Should these transformations have little effect and parametric modes of analysis are not valid, non-parametric modes, such as a Mann–Whitney U test, were used. Only violations of these assumptions have been reported. Four result categories are provided: (i) Phase 1, the route following task; (ii) Phase 2, the map drawing task; (iii) Phase 3, the landmark locating task; and (iv) Phase 4, correlations between participant task-performance and SBSOD scores.

4.1. Phase 1: Route Following Task

To test hypothesis 1 predicting that the LM Group would outperform the GM Group in regard to accurately following the route, the mean number of errors was calculated for the LM and GM participant groups (LM: M = 0.8, SD = 1.03; GM: M = 1.1, SD = 0.74). Clearly, these figures were too low to warrant a formal error analysis, and the differences between the participant groups were minimal. Hypothesis 1 was therefore refuted.

For the PM Group, failure to adhere to the route described to the researcher prior to embarking on the walk resulted in the errors recorded. In regard to errors, five participants in the LM Group, two in the GM Group, and eight in the PM Group made no errors at all. Only one hesitation was recorded in the PM Group, with no hesitations recorded with the other groups. Seven of the ten PM Group participants chose the simplest route to the destination with only two turns. Regarding the walking times, two outliers were found in the LM Group's data, as assessed by inspection of a boxplot. These two scores were then adjusted to be slightly above the next highest and below the next lowest scores, a transformation process described by Laerd Statistics [30]. All data were normally distributed; however, the assumption of homogeneity of variances was violated, as assessed by Levene's test for equality of variances ($p = 0.017$). Therefore, a one-way Welch ANOVA was computed on these means. The walking times ranged from 18:28 min to 26:50 min, with an almost identical mean travel time for the LM and GM Groups (LM Group: M = 22:05 min, SD = 2:56 min; GM Group: M = 22:44 min, SD = 1:07 min) and a slightly shorter time for the PM Group (M = 21:30 min, SD = 2:26 min). These means are shown in Figure 11.

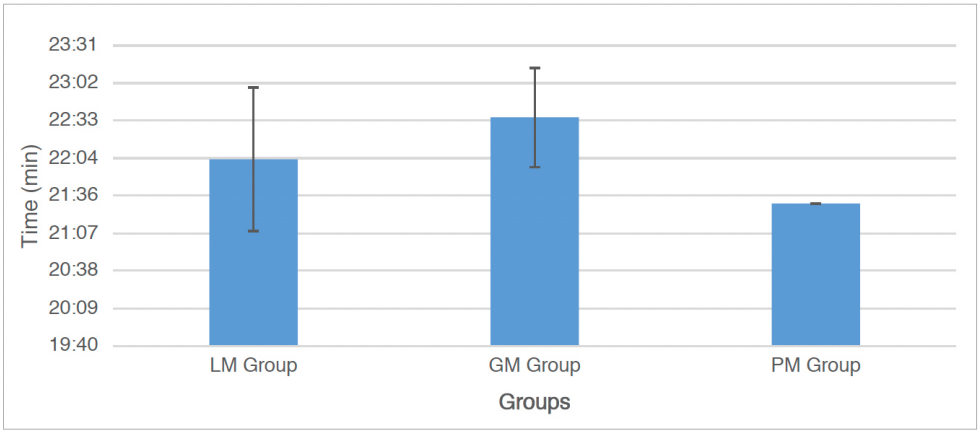


Figure 11. Mean walking times in Phase 1. Error bars indicate Standard Errors (SEs).

There were no statistically significant differences in walking time between the three participant groups (Welch's F(2, 15.187) = 1.1, $p = 0.358$). It can therefore be concluded that the type of navigational aid guiding the walk did not significantly influence participants' walking time.

4.2. Phase 2: Route Drawing Task

Hypothesis 2 predicted that the GM Group would draw the route more accurately than the LM Group on an unlabeled paper map after walking the route. Accuracy was determined by the number of erroneously drawn route segments (between each direction point). As shown in Figure 12, the mean drawing scores for these groups varied somewhat (LM: M = 2.9, SD = 2.4; GM: M = 0.6, SD = 1.35). Two outliers were found and excluded, and the assumption of normality was violated, as indicated by the Shapiro–Wilk test ($p \leq 0.05$). Consequently, a Mann–Whitney U test was conducted. The descriptive statistics for the Mann–Whitney U test provide an overview of the data related to the number of erroneously drawn route segments. The participants (LM and GM Groups) made an average of 1.22 errors when drawing the route. The errors varied widely, ranging from 0 to 5, with a standard deviation of 1.59, indicating considerable variability in participants'

performance. The mean rank for incorrectly drawn route segments was 12.13 for the LM Group and 7.40 for the GM Group, suggesting that the LM Group generally made more errors in drawing the route than the GM Group. The Mann–Whitney U-value is 19.000, with a Z-value of −2.074. The asymptotic significance (p-value) is 0.038, which is less than 0.05. This indicates a statistically significant difference between the two groups in terms of the number of incorrectly drawn route segments. The LM Group had a higher median number of errors (2.00) compared to the GM Group, which had no errors (0.00) using an exact sampling distribution for U [31]. This median comparison reinforces the earlier finding that the LM Group made more errors than the GM Group. The results suggest that participants in the GM Group performed better than those in the LM Group in terms of accurately drawing the route, as indicated by both the mean ranks and the median number of errors, thus supporting hypothesis 2. The r-value is approximately 0.49, indicating a medium effect size.

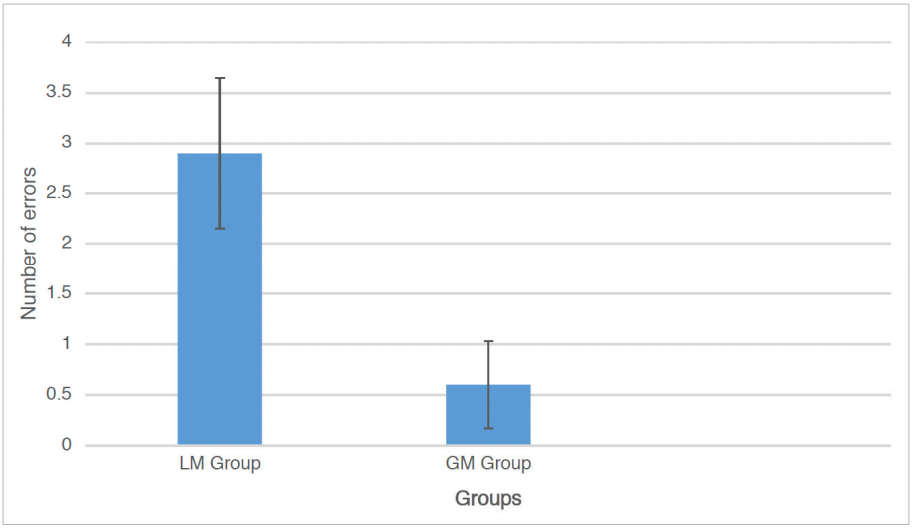


Figure 12. Errors made when drawing the route in Phase 2. Error bars indicate Standard Errors (SEs). The PM Group did not walk the same route.

4.3. Phase 3: Locating Landmarks Task

To test hypothesis 3, which anticipated that the LM Group would correctly identify more landmarks on the unlabeled map compared to the GM and PM Groups, a one-way ANOVA was conducted. The analysis used the number of correctly located landmarks as the dependent variable, with the independent variable being the group category, consisting of three levels: LM, GM, and PM. The mean values and standard deviations for the groups were as follows: LM: M = 7.1, SD = 3.51; GM: M = 2.6, SD = 1.96; PM: M = 1.9, SD = 1.79, as shown in Figure 13. This indicates that the LM Group had the highest average number of correctly located landmarks, followed by the GM Group, with the PM Group having the lowest average.

The ANOVA revealed a significant difference between the groups ($F(2, 27) = 12.343$, $p < 0.001$), with a large effect size ($\eta^2 = 0.478$), suggesting that group membership accounted for a considerable portion of the variance in landmark placement accuracy. Post hoc tests (Tukey HSD) confirmed that the LM Group was significantly more accurate in locating landmarks than both the GM and PM Groups, while no significant difference was found between the GM and PM Groups.

These findings strongly support hypothesis 3, confirming that the LM Group was significantly better at correctly placing landmarks compared to the other two groups. The results imply that the LM Group's approach or experience was more effective in landmark recognition and placement on the map. Hypothesis 3 was therefore confirmed.

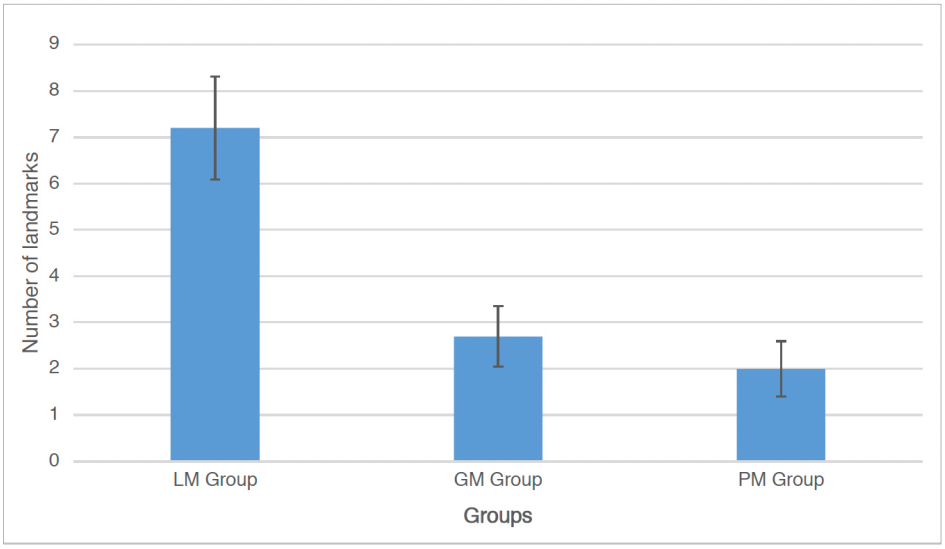


Figure 13. Landmarks correctly located by each of the three participant groups in Phase 2. Error bars indicate Standard Errors (SEs).

Next, in order to test hypotheses 4 and 5, the data were divided into on- or off-route landmarks. Dealing with hypothesis 4 first, according to which all participant groups were predicted to locate more on-route landmarks correctly than off-route landmarks. Figure 14 shows this to be true. (LM: M = 5.5, SD = 2.55; GM: M = 2.0, SD = 1.33; PM: M = 1.1, SD = 0.99). Consistent with the prediction of hypothesis 4, the mixed-design 3 x (2) ANOVA computed for the three participant groups (LM, GM, PM) and landmarks (on-route, off-route) revealed a main effect for landmarks (F (1, 27) = 46.00, $p < 0.001$). Hypothesis 4 was therefore supported.

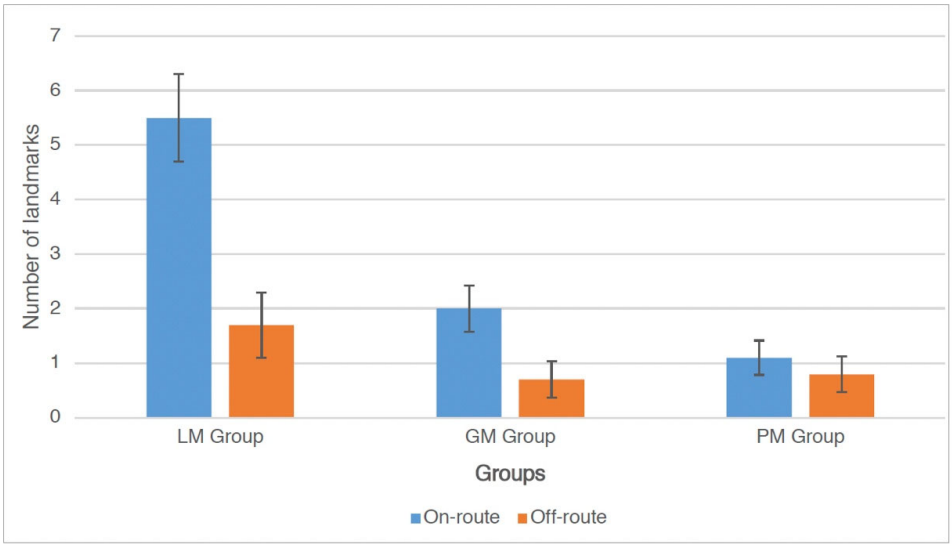


Figure 14. Both types of landmarks correctly located by the three groups. Error bars indicate Standard Errors (SEs).

Although not specifically hypothesized, a comparison of the data for the on-and off-route landmarks within each of the three groups yielded a main effect for groups as well ($F(2, 27) = 15.56$, $p < 0.001$). In addition, the interaction was also significant ($F(2, 27) = 15.56$, $p < 0.001$). To determine the locus of the interaction, simple main effects showed that the data for on-route landmarks differed significantly between all three participant groups

($p < 0.001$). Significant differences also occurred with the locating of on-route landmarks compared to off-route landmarks (LM: $p < 0.001$; GM: $p = 0.004$).

With respect to hypothesis 5 predicting that participants in the LM Group would locate more off-route landmarks correctly than participants in the two other groups, Figure 14 above suggests that to be true (LM: M = 1.7, SD = 1.88; GM: M = 0.7, SD = 1.06; PM: M = 0.8, SD = 1.03). However, the analysis of those data was not as simple as for the on-route landmarks. First, four statistical outliers emerged in the off-route landmark data assessed by inspection of a boxplot. These were therefore reduced downwards. The data were also not normally distributed, as assessed by Shapiro–Wilk's test of normality ($p > 0.05$), and a data transformation failed to rectify this situation. However, as ANOVAs are generally resilient to deviations from normality [30], the aforementioned mixed-design ANOVA was still carried out. Homogeneity of variances was violated ($p < 0.05$), as assessed by Levene's test, yet there was homogeneity of covariances ($p = 0.09$) as described by Box's M test. The interaction was highly significant, F(2, 27) = 15.56, $p < 0.001$, partial 2 = 0.54. To determine the locus of the interaction, simple main effects were calculated for both factors. Contrary to the prediction of hypothesis 5, the analysis for the off-route landmarks yielded no difference between the three groups (LM, GM, and PM), F(2, 27) = 1.086, $p = 0.352$, partial 2 = 0.074, $p = 0.352$. Hypothesis 5 was therefore rejected.

The data plot in Figure 15 shows the number of correct placements of the on-route landmarks by their sequential location in the environment. It suggests the presence of a primary and a recency effect, most pronounced for the LM Group. The primacy effect is the effective recall of a particular stimulus first experienced; the recency effect is the effective recall of a particular stimulus last experienced [32]. Whereas the data for the other two participant groups also suggest a weak primacy effect, the recency effect is largely absent for the GM and the PM Groups. This suggests that, once walking the route, participants in these two groups relied on their navigational aids to follow the route, with no need to refer to landmarks, unlike the LM Group, who relied upon them exclusively. This is further discussed in the Section 5.

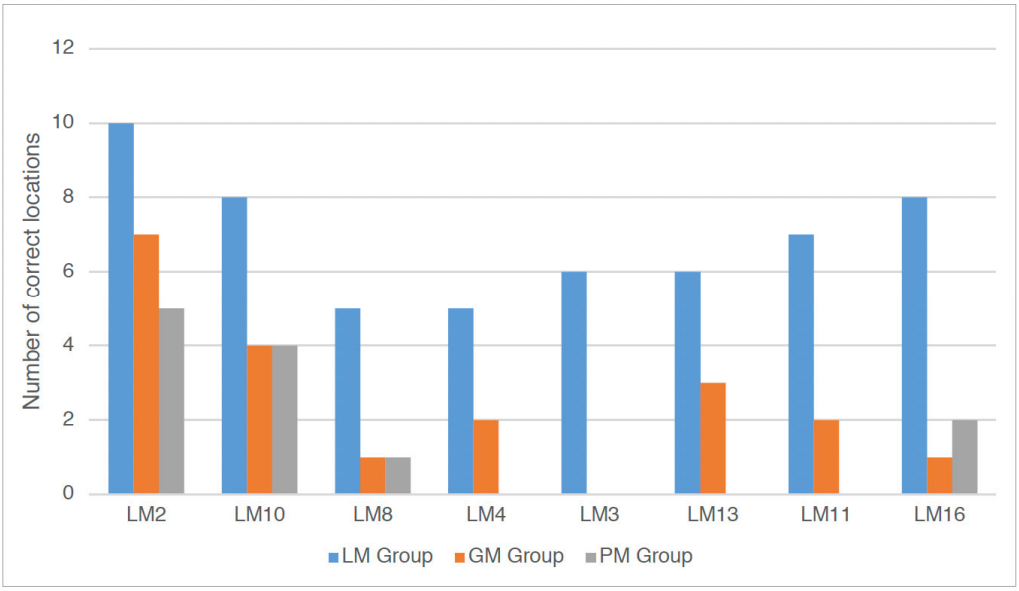


Figure 15. The total number of on-route landmarks correctly located in the sequence encountered on the route.

4.4. Phase 4: The SBSOD Scale

The mean SBSOD scores calculated for the three participant groups were quite similar: (LM: M = 3.81/7, SD = 1.33; GM: M = 4.12/7, SD = 0.90; PM: M = 3.21/7, SD = 1.47). Pearson product–moment correlation coefficients were computed to determine the relationship between SBSOD scores and performance for each participant group in Phase 1, which was

the only phase involving wayfinding through environmental space. Examination of the number of route-following errors and hesitations in Phase 1 did not reveal any significant correlations either. One might have expected that the PM Group's scores could yield a correlation, as map-based conditions in other studies have shown correlations, which is further clarified in the next section.

5. Discussion

Hypothesis 1, predicting that the LM Group would outperform the GM Group in regard to accurate route-following, was clearly not supported: all participants in the LM and GM Groups were able to follow the routes described with relative ease, making few errors and hesitations. Even though the route length and complexity for the LM and GM Groups were modeled on previous research [2,9,10], the route was evidently too easy to allow discrimination in the performance of the two groups. To rectify this, a longer and more complex route should be used in subsequent studies to observe more errors and hesitations when route-following. The PM Group generally chose a simple route with two turns, which was easily followed.

According to the confirmed hypothesis 2, the GM Group was more successful at drawing the route on an unlabeled map than the LM Group. In fact, almost all participants in the GM Group were able to draw their routes 100% accurately, resulting in extremely few errors and also confirming that the route was likely too short and easy to follow. Participants in the LM Group drew wrong turns more frequently, and their drawing errors were mainly connected with the large park encountered along the route where three turns were required to navigate around it. A possible reason for this behavior is that the attention of the LM Group when following the route was focused on landmarks, which were predominantly conspicuous buildings (singular points in the region). But as the park was also a landmark (an area in the region), perhaps too much time was spent focused on the park as a landmark when wayfinding rather than on the routes around the park. This also suggests that large area-based landmarks may not perform as well for wayfinding as more point-like landmarks. It is also interesting that the LM Group were able to draw the route—certainly not as accurately as the GM Group—even though they used no map depicting the route. As the PM Group walked different routes, a direct comparison was not possible, but it can be noted that the common two-turn route selected by this group was drawn with almost 100% accuracy. Avoiding complexity, the simplest route will always be the easiest to draw. Another factor contributing to the high performance of the route drawing may have been the scale of the map used. The route sat comfortably in the center of the map area, meaning that participants did not have to work too hard to locate the route area on the map before drawing the route. It appears the pre-set route was too easy to follow, but the length of the route and the number of turns used are close to the mean number of those attributes in similar wayfinding studies [2,9,10,33].

Regarding locating landmarks, the LM Group located more total landmarks correctly than participants in the GM and PM groups, confirming hypothesis 3, and all three groups located more on-route landmarks than off-route landmarks, confirming hypothesis 4. This makes sense, as the LM and GM Groups followed the same route. Consistent with traditional memory studies, the primacy effect for the LM Group provides evidence that most attention was paid to the first two on-route landmarks and, to a lesser extent, to the last two on-route landmarks, thereby resulting in a weak recency effect as well. Once the first few items (actual landmarks) had been identified by comparing these with the photographs shown on the navigation aid, such comparisons were all they needed to reassure themselves that they were on the right track. With 16 landmarks thus shown, fewer attentional resources were available, although the recent effect suggests that participants still attempted to commit all landmarks to memory. Alas, only the last two stood out, as these were fresh in the participants' memories. It is also noteworthy that the first and last on-route landmarks were the largest used in this study, which might have further enhanced both the primacy and the recency effects.

The observation that the LM Group also located landmarks on a map that did not indicate the route per se suggests that those participants acquired more than sheer route knowledge, namely an initial level of survey knowledge. This supports the contention that landmark knowledge also contributes to the development of survey knowledge. It is tempting to interpret the observation that the LM Group both recalled more on-route landmarks and were able to place more of these correctly than participants in the other two groups that they had engaged in active encoding of landmark information while walking the route. As they were shown the landmarks again in Phase 2, the results equally suggest it was a case of cued recall.

It is interesting to note that there was some evidence of a primacy effect but not a recent effect on the performance of the GM Group. One way to account for that could be because their navigation tool did not include images of landmarks. Attention would therefore most likely have been focused on following the map-based route correctly. At the start of the route, some GM Group participants were observed visually matching their maps to the environment, with the first couple of nearby on-route landmarks likely noticed. Once progress was satisfactory, participants could trust the map without paying attention to the surroundings. Regarding the PM Group, as many of these participants generally walked routes away from the main route, the on-route landmarks found along this route were not noticed.

The eight off-route landmarks were all visible in the region, but relatively few of these were correctly located on the maps. It is perhaps not surprising that the four off-route landmarks situated nearest to the main route were also those located correctly most often. They were situated 100–150 m from the route, while three of the other four were highly noticeable tall office towers, and another, less noticeable large building, was situated between approximately 300 and 500 m from the route. This suggests that the further a landmark is located from a given path/route, the less likely it will be detected and subsequently recalled.

The refuted hypothesis 5 claimed that because explicit attention was drawn to landmarks, the LM Group would locate more off-route landmarks than the other groups. One possibility is that the route was so simple and the on-route landmarks so obvious that the LM participants had no need to attend to off-route landmarks. Had the route been more complicated, perhaps participants would have needed to rely more on orientation to help them maintain their bearings and understand the space they were traveling through. This is when off-route landmarks may have been noticed as they aid orientation.

As indicated by their mean SBSOD scores, all participant groups rated themselves around average in regard to their perceived environmental spatial abilities. Overall, the scale is most closely related to tasks that involve movement through environmental space where spatial updating is necessary. Research in which correlations have been found has typically involved orientation/direction pointing, route estimation tasks, and only sometimes route-following, as conducted here. Studies reporting correlations between the SBSOD scale and mobile maps in the outdoor wayfinding context are still very scarce. To the best of the authors knowledge, no research has been published in regard to wayfinding solely with sequential images of properly classified landmarks and the SBSOD scale as used by the LM Group here. As Phase 1 directly involved spatial updating, a correlation was predicted in this phase, in particular with the paper map condition given the paucity of correlations with mobile maps. However, the fact that no correlation was observed in that phase suggests that accurately assessing one's own spatial ability may not be a straightforward task. The route may also have been too easy to follow to truly test participants' spatial abilities. Alternatively, repeating the route-walking phase may have increased spatial knowledge of the region, thereby demonstrating a stronger correlation with SBSOD scores. The relatively small sample size of 30 participants may have impacted the ability to find correlations here; however, group sizes of 10 are usually adequate to result in statistically reliable group differences.

5.1. Implications of the Research

The research on using photographs of landmarks as navigational aids for pedestrians has the following significant implications:

5.1.1. Enhanced Wayfinding Strategies

This study demonstrates that incorporating landmark imagery into navigation aids can significantly improve wayfinding performance. Participants who used photographs of landmarks made fewer navigation errors and hesitations compared to those relying on traditional map-based systems. This suggests that landmarks serve as effective cognitive tools that enhance spatial awareness and route-following capabilities, particularly in complex pedestrian environments.

5.1.2. Mobile Mapping Applications

The findings advocate for the integration of landmark-based guidance in mobile mapping applications. Current mobile maps do not use images of landmarks in their interfaces. By incorporating landmarks, mobile maps could provide users with more intuitive and contextually relevant navigation experiences, potentially addressing issues like the "keyhole problem" and reducing reliance on automated systems that may fail.

5.1.3. Cognitive Engagement in Navigation

The research highlights the cognitive processes involved in wayfinding, particularly the differences between using landmark-based navigation and other navigation aids. This study suggests that engaging with landmarks requires users to actively participate in their navigation, fostering a better understanding of their surroundings. This contrasts with the passive reliance on automated directions, which can diminish spatial knowledge and awareness.

5.1.4. Future Research Directions

This study opens avenues for further research into the role of landmarks in navigation. Future investigations could explore travel along longer, more complicated routes, how to best use landmarks in map interfaces, and the potential for augmented reality applications that visualize landmarks in real time.

6. Conclusions

This study demonstrated that—at least when needing to travel just brief distances along a predetermined route—landmarks can act as useful guides for pedestrians traversing a new environment. Participants in the LM and GM Groups were thus able to efficiently follow a simple 1.7 km route. A key finding here is that the LM Group acquired more initial survey knowledge than the GM Group and PM Group participants as their navigation aid attuned them to landmarks on the route, enabling them to show more of them on a map that did not show the route. While landmarks thus facilitated wayfinding and recall of the precise location of the landmarks encountered, the LM Group participants' ability to draw the route just walked accurately was attenuated compared with the GM and PM Groups. The results from the PM Group in this study were similar in general to those obtained by the GM Group. It is concluded that incorporating landmarks into contemporary navigation aids such as mobile maps should be further investigated as landmarks may be able to work in conjunction with the maps and routes generated in mobile maps to facilitate pedestrian wayfinding.

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Data Availability Statement: The data presented in this study is available on request from the corresponding author.

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